Reinforcement Learning Control for 6 DOF Flight of Fixed-Wing Aircraft

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Abstract: Flight control is a key technique for the autonomous unmanned aircraft. The traditional model-based controller design approaches often aim at concrete plant and are short in universality. Reinforcement learning provides a general controller design paradigm that is adaptive, optimized, model-free and widely applicable, and it is a promising way for the intelligent control. In contrast to the 3 Degree-of-freedom (DOF) flight, the 6 DOF motion better describes the aircraft real flight, while the implementation of the intelligent control is much harder. Based on the multiple continuous states input and multiple continuous action output deep reinforcement learning, the integrated intelligent control for the cruise flight, directly from the vehicle flight states to the aero-surfaces and thrust control, is developed for the full-sized fixed-wing aircraft. It avoids the artificial trajectory and attitude loop separation in the traditional controller design, and is advantageous to the exploitation of the aerodynamics and the nonlinear inertia coupling.

Key Words: fixed-wing aircraft; flight control; artificial intelligence; intelligent control; deep reinforcement learning

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1 INTRODUCTION

Flight control is a key technique for the development of autonomous unmanned aircraft system. It includes the take-off and landing control, attitude hold, height control, and trajectory tracking. The PID control method is widely used for the aircraft flight control. However, due to the nonlinearity of the aircraft dynamics, the stability of the linear controller cannot be guaranteed theoretically. Moreover, the design procedure is time-consuming since it involves multiple trim points selection and control gain tuning [1]. The modern control technologies, including the LQR control [1], the dynamic inverse control [2], the backstepping control [3], the sliding mode control [4], and the L1 adaptive control [5], etc., have also been extensively studied for the aircraft flight control. In general, these control schemes achieve good performance. Nevertheless, the PID control and modern control methods are model-based. In practice, there always exist errors and disturbances for the plant modeling. For the complex system like the aircraft, which contains mechanics, electricity, and structure sub-systems, although various robust control approaches are proposed, the potential poorly-understood uncertainties, such as the nonlinear unsteady aerodynamics at high angle-of-attack [6], may ruin the control effect or even bring disaster.

Artificial Intelligence (AI) may greatly develop the human productivity and bring historical changes into the human society. Learning is an essential feature of intelligence and currently Machine Learning (ML) is the most promising branch in the field of AI research, which may be classified as the supervised learning, unsupervised learning and reinforcement learning [7]. Among them, reinforcement learning is widely applicable for the dynamic decision-making problems. Based on such approach, agents can interact with the environment and learn behavioral strategies that maximize specific index [8]. After years of development, the value function-based reinforcement learning method and the direct policy search reinforcement learning method are proposed upon the theory of dynamic programming, and the Actor-Critic methods that integrate the two methods are formed [9-12]. Compared with the traditional control methods, reinforcement learning provides a general controller design paradigm that is adaptive, optimized, model-free, and widely applicable, and is a promising way to realize the intelligent control.

Neural Network (NN) is an information processing system which imitates the structure and function of human brain. It can approximate arbitrary nonlinear function and is an effective modeling tool [13]. In particular, deep NN model can learn complex functions and produce better performance, through refining features among hidden layers [14]. Deep Reinforcement Learning (DRL) is the variation of reinforcement learning that employs the deep NN model. The Deep Q Network (DQN) algorithm is a well-known DRL method, which achieves the level of human players in Atari games. However, it can only solve problems with “discrete action” [15]. To address such issue, the Deep Deterministic Policy Gradient (DDPG) algorithm is further developed for the robot motion control [16], and is now a popular algorithm for the "continuous state, continuous action" problems. With the DDPG algorithm, the intelligent control of the quadrotor hover and forward flight is realized in Ref. [17].

On the other hand, the control of the aircraft flight often separate the plant into the outer-loop trajectory control and the inner-loop attitude control [18]. This ruins the performance of the aircraft since the nonlinearities are not exploited. The reinforcement learning control can take advantages of the coupling effect between the trajectory and the attitude dynamics and achieve optimized integrated

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flight control. In Ref. [19], the intelligent control for two different aerobatic maneuvers of the fixed-wing aircraft are developed by one variant of DQN, and it well explores and exploits the nonlinearities of dynamics. In this paper, the DDPG algorithm is employed to develop the 6 Degree-of-freedom (DOF) cruise controller for a full-sized fixed-wing aircraft. By directly learning the control law from the flight states to the aero-surfaces and thrust control, the intelligent integrated flight controller is developed and it avoids the separation of guidance and control, which is common during the traditional controller design.

2 MATHEMATICAL MODEL

The fixed-wing aircraft body coordinate frame *b* and the ground coordinate frame *g* are defined first [20]. The body frame *b* is fixed with the aircraft, with its origin o_b located at the center of mass. The $o_b x_b$ axis lies in the longitudinal symmetrical plane and points to the nose. The $o_b y_b$ axis is perpendicular to the symmetry plane and points to the right of the fuselage. The $o_b z_b$ axis lies in the symmetry plane and points down. The ground system *g* has the origin o_g that is located at some point on the ground. The $o_g x_g$ and $o_g y_g$ axes lie in the horizontal plane, pointing to the north and the east, respectively. The $o_g z_g$ axis is parallel to the vertical direction, towards the center of the earth.

Assume the thrust is along the $o_b x_b$ axis, the kinematic and dynamic equations for the aircraft trajectory motion are

$$\dot{\boldsymbol{r}} = \begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{z} \end{bmatrix} = \boldsymbol{R}_b^g \begin{bmatrix} u \\ v \\ w \end{bmatrix} \tag{1}$$

$$\dot{\boldsymbol{v}} = \begin{bmatrix} \dot{u} \\ \dot{v} \\ \dot{w} \end{bmatrix} = \begin{bmatrix} vr - wq - gR_g^b(1,3) + \dfrac{X + \eta T_{\max}}{m} \\ -ur + wp - gR_g^b(2,3) + \dfrac{Y}{m} \\ uq - vp - gR_g^b(3,3) + \dfrac{Z}{m} \end{bmatrix} \tag{2}$$

where $\boldsymbol{r} = \begin{bmatrix} x & y & z \end{bmatrix}^{\mathrm{T}}$ is the aircraft position vector in the ground reference frame, with "T" denoting the transpose operator, and $\boldsymbol{v} = \begin{bmatrix} u & v & w \end{bmatrix}^{\mathrm{T}}$ is the aircraft velocity vector, described in the body frame. m is the mass. X, Y, and Z are the aerodynamic force, along the $o_b x_b$, $o_b y_b$, and $o_b z_b$ axis, respectively. η is the throttle of engine. g is the acceleration of gravity. $\boldsymbol{R}_b^g$ is the rotation matrix from the body frame *b* to the ground frame *g*.

The aircraft attitude kinematic and dynamic equations are

$$\dot{\boldsymbol{q}} = \frac{1}{2} \begin{bmatrix} -q_1 & -q_2 & -q_3 \\ q_0 & -q_3 & q_2 \\ q_3 & q_0 & -q_1 \\ -q_2 & q_1 & q_0 \end{bmatrix} \boldsymbol{\omega} \tag{3}$$

$$\dot{\boldsymbol{\omega}} = \begin{bmatrix} \dot{p} \\ \dot{q} \\ \dot{r} \end{bmatrix} = \boldsymbol{I}^{-1} \left\{ -\begin{bmatrix} 0 & -r & q \\ r & 0 & -p \\ -q & p & 0 \end{bmatrix} \boldsymbol{I} \begin{bmatrix} p \\ q \\ r \end{bmatrix} + \begin{bmatrix} l_a \\ m_a \\ n_a \end{bmatrix} \right\} \tag{4}$$

where $\boldsymbol{q} = \begin{bmatrix} q_0 & q_1 & q_2 & q_3 \end{bmatrix}^{\mathrm{T}}$ is the attitude quaternion and $\boldsymbol{\omega} = \begin{bmatrix} p & q & r \end{bmatrix}^{\mathrm{T}}$ is the angular velocity. $\boldsymbol{I}$ is the inertia matrix of the aircraft. l_a, m_a and n_a are the aerodynamic moments, along the $o_b x_b$, $o_b y_b$ and $o_b z_b$ axis, respectively.

The aircraft considered in the following is a full-sized F/A-18 aircraft. The mass is $m = 15119\text{kg}$. The inertia components are I_{xx} =31184kg·m^2, I_{yy} =205125kg·m^2, I_{zz} =230414 kg·m^2, and I_{xz} =-4028.1kg·m^2. The aerodynamics are given in Ref. [21].

3 MODELING AND VERIFICATION OF NN CONTROLLER

In order to investigate the generalization performance of the NN controller model, first the PID flight control law is designed and used to develop the NN controller through the supervised learning.

3.1 PID Controller Design

To realize the cruise flight of the aircraft, the longitudinal and lateral control laws are designed. The longitudinal control includes the speed control and the height control. The speed control law is

$$\eta = -k_{pV}(V - V_{\text{cmd}}) - k_{iV}\int (V - V_{\text{cmd}})\mathrm{d}t \tag{5}$$

where V_{cmd} is the commanded aircraft speed and $V = \sqrt{u^2 + v^2 + w^2}$ is the aircraft speed. k_{pV} and k_{iV} are the corresponding proportional and integral control gains. The height control law is

$$\delta_e = -k_q q - k_\theta(\theta - \theta_{\text{cmd}}) \tag{6}$$

where δ_e is the elevator. k_q and k_θ are the proportional control gains for the pitch rate q and the pitch angle θ. θ_{cmd} is the pitch angle command and is given by

$$\theta_{\text{cmd}} = -k_h(h - h_{\text{cmd}}) - k_{V_z} V_z - k_{ih}\int (h - h_{\text{cmd}})\mathrm{d}t \tag{7}$$

where $h = -z$ is the height and h_{cmd} is the commanded height. V_z is the aircraft longitudinal vertical speed. k_h and k_{V_z} are the proportional gains regarding the height and the speed. k_{ih} is the integral gain of the height control.

The lateral-directional control laws are

$$\delta_a = k_p p + k_\phi(\phi - \phi_{\text{cmd}}) + k_{i\phi}\int (\phi - \phi_{\text{cmd}})\mathrm{d}t \tag{8}$$

$$\delta_r = k_r r - k_\beta \beta \tag{9}$$

where ϕ is the aircraft roll angle and β is the aircraft sideslip angle. k_p, k_ϕ, k_r, and k_β are the proportional control gains for the corresponding variables. $k_{i\phi}$ is the integral control gain. The roll angle command ϕ_{cmd} is

$$\phi_{\text{cmd}} = -k_\psi(\psi - \psi_{\text{cmd}}) \tag{10}$$

where ψ is the yaw angle, and ψ_{cmd} is the commanded yaw angle, which represents the desired heading when there is no sideslip. k_ψ is the proportional control gain.

3.2 NN Controller Training Through Supervised Learning

The observations for the aircraft flight control are given by a one-dimensional vector, and the NN model with fully connected feedforward multi-hidden layers is suitable for the modeling of the controller. In order to exam the generalization ability of the NN model, the supervised learning is implemented. With Pytorch, the developed NN controller model has 5 layers. The input layer has 10 input units, including the height, the velocity vector, the Euler angles (transformed from the quaternion) and the angular velocity vector. The output layer has 4 output units, corresponding to the aileron, elevator, rudder and throttle. There are 3 hidden layers in the NN model, and the number of units are set as 32, 64 and 128, respectively. The first 4 layers use “Relu” activation function while the output layer uses “Tanh” activation function.

Take the PID controller as "teacher"; the NN controller is trained through the labeled data generated in flight simulations with different initial conditions. The network parameters are initialized by the “xavier_uniform” function, and the “Adam” method is used for the optimization [22]. In the supervised learning, the learning rate is 1×10^{-3}, the batch size is 128, and the number of training iteration is 20000. The trained NN controller is applied to the flight simulation. Starting from the origin of the ground frame, the aircraft has an initial speed of 100m/s and the initial heading angle is 0^{o}. The commanded height is 100m. The commanded speed is 76.32m/s and the commanded heading angle is 31.61^{o}. Simulation results under the PID controller and the NN controller are presented in Figs. 1-3. Figure 1 plots the height curve of the aircraft and Fig. 2 gives the velocity profiles. In Fig. 3, the control commands are compared. From these figures, it is shown that the performance of the NN controller is very similar to that of the PID controller. This proves that the NN model meets the modeling requirement for the intelligent controller design.

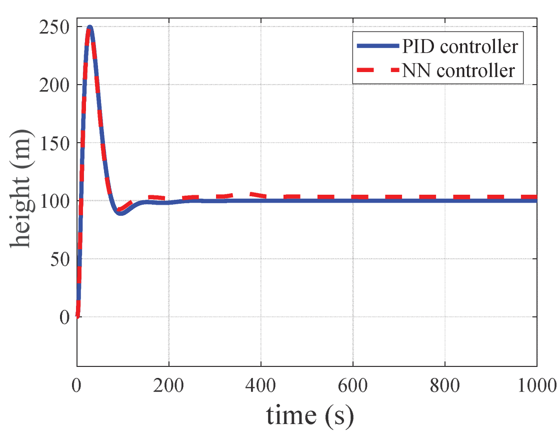

Fig. 1 Height curves under the PID controller and the NN controller.

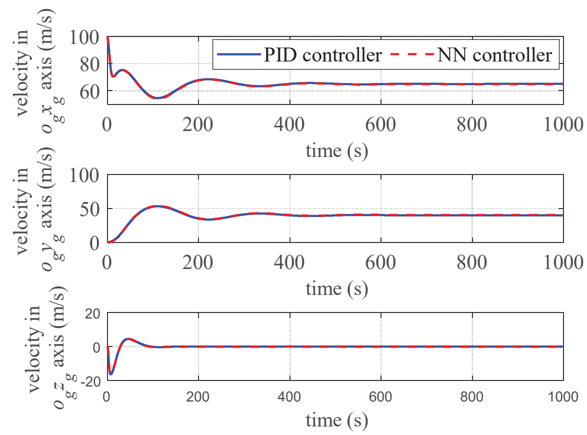

Fig. 2 Velocity profiles under the PID controller and the NN controller.

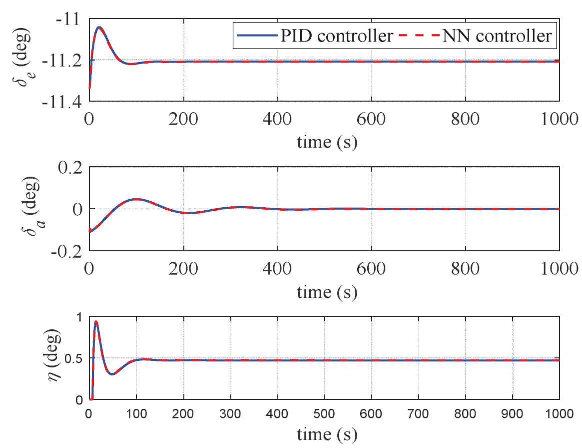

Fig. 3 Control profiles under the PID controller and the NN controller.

4 NN CONTROLLER DESIGN BY REINFORCEMENT LEARNING

With the improvement of algorithms and the development of hardware capability, the training of deep NN models is no longer the bottleneck that restricts the studies on deep (reinforcement) learning [23]. Thus, in the following the NN controller is trained through directly interacting with the environment from a random initialization of NN parameters. By exploring the simulation data, the intelligent controller may exploit the coupling effect of the nonlinear dynamics and achieve end-to-end integrated optimal control.

4.1 DDPG Algorithm

The DDPG algorithm is developed to solve the "continuous action" type problems, upon the DQN algorithm. It uses the Actor-Critic architecture, where the Critic evaluates the action value function of the policy, i.e., $Q = Q(\boldsymbol{x},\boldsymbol{u})$, and the Actor defines a deterministic policy function $\boldsymbol{u} = \pi(\boldsymbol{x})$. With the NN model to approximate these functions, the Critic NN is expressed by $\tilde{Q}(\boldsymbol{x},\boldsymbol{u};\boldsymbol{\theta}_Q)$, with $\boldsymbol{\theta}_Q$ denoting the parameters of the Critic NN; the Actor NN is expressed by $\tilde{\pi}(\boldsymbol{x};\boldsymbol{\theta}_u)$, with $\boldsymbol{\theta}_u$ the NN parameters. Since the agent may learn the optimal policy regardless of the employed policy, the DDPG algorithm falls into the category of the off-policy method. The update law of the Critic NN parameters is similar to that of DQN, established by reducing the following performance index

$$J = (\Delta Q)^2 = \left(y_t - \tilde{Q}(\boldsymbol{x}_t, \boldsymbol{u}_t; \boldsymbol{\theta}_Q)\right)^2 \tag{11}$$

where y_t is the estimation of the action value, $\tilde{Q}(\boldsymbol{x}_t, \boldsymbol{u}_t; \boldsymbol{\theta}_Q)$ is the prediction of the action value, and ΔQ is their difference. Hence the update law for the Critic NN parameters is

$$(\boldsymbol{\theta}_Q)_{k+1} = (\boldsymbol{\theta}_Q)_k + \alpha \Delta Q \frac{\partial Q}{\partial (\boldsymbol{\theta}_Q)_k} \tag{12}$$

where α is the learning rate of the Critic network. On the other hand, in order to make the outputs of the Actor network maximize the output of the Critic network, the gradient of the action value function with respect to the Actor network parameters may be derived as $\frac{\partial Q}{\partial u}\frac{\partial u}{\partial \boldsymbol{\theta}_u}$, and then the update law for the Actor network parameters is

$$(\boldsymbol{\theta}_u)_{k+1} = (\boldsymbol{\theta}_u)_k + \beta \frac{\partial Q}{\partial u}\frac{\partial u}{\partial (\boldsymbol{\theta}_u)_k} \tag{13}$$

where β is the learning rate of the Actor network.

Like the DQN, the DDPG algorithm also considers the target networks that make the training stable, including the Target-Critic network $\tilde{Q}(\boldsymbol{x}, \boldsymbol{u}; \boldsymbol{\theta}_Q^{'})$ and the Target-Actor network $\tilde{\pi}(\boldsymbol{x}; \boldsymbol{\theta}_u^{'})$. They have the same structure to the original networks, and the corresponding parameters are denoted by $\boldsymbol{\theta}_Q^{'}$ and $\boldsymbol{\theta}_u^{'}$, respectively. With the target networks, reinforcement learning behaves similarly to the supervised learning, and more stable learning process may be achieved. This is beneficial to improve the convergence of NN parameters. The target network parameters adopt a gradual manner update law, that is

$$\boldsymbol{\theta}_Q^{'} = \tau \boldsymbol{\theta}_Q + (1-\tau)\boldsymbol{\theta}_Q^{'} \tag{14}$$

$$\boldsymbol{\theta}_u^{'} = \tau \boldsymbol{\theta}_u + (1-\tau)\boldsymbol{\theta}_u^{'} \tag{15}$$

where τ is the update rate. Obviously the larger τ is, the faster the update will be. In addition, the DDPG algorithm also introduces the experience pool, which is advantageous to attenuate the correlation of data and address the non-stationary distribution of samples through mini-batch sampling. Consider the target networks and the mini-batch sampling, Eq. (11) needs to be adapted as

$$J = \sum_N (\Delta Q)^2 = \sum_N \left(y_t - \tilde{Q}(\boldsymbol{x}_t, \boldsymbol{u}_t; \boldsymbol{\theta}_Q)\right)^2 \tag{16}$$

where N is the size of the mini-batch. The action value function is estimated as

$$y_t = r_t + \gamma \tilde{Q}(\boldsymbol{x}_{t+1}, \boldsymbol{u}_{t+1}; \boldsymbol{\theta}_Q^{'}) \tag{17}$$

where r_t is the reward function and γ is the discount factor. Correspondingly, the gradients on the parameters in Eqs. (12) and (13) need to be modified as the sum on the whole mini-batch data set.

In particular, in order to balance the exploration and exploitation mechanism, the control output by the Actor network will be blended with random noise during the learning.

4.2 NN Controller Learning

For the cruise control of the fixed-wing aircraft, in the reinforcement learning the commanded height is h_c, while the commanded velocity is given in the ground frame, namely, $\begin{bmatrix} v_{xc} & v_{yc} & 0 \end{bmatrix}^{\mathrm{T}} = [V_{\mathrm{cmd}}\cos\psi_{\mathrm{cmd}} \quad V_{\mathrm{cmd}}\sin\psi_{\mathrm{cmd}} \quad 0]^{\mathrm{T}}$. Analogous to the integral term in the PID controller that eliminates the steady-state error, the Actor network has 13 inputs; they are the height error, the three-dimensional velocity error, the Euler angles, the angular velocity, the integral of height error, and the integrals of velocity error along the $o_g x_g$ and $o_g y_g$ axes. The 4 action outputs are aileron, elevator, rudder and throttle, respectively. For the Critic network, there are 17 inputs, including the 13 state inputs of the Actor network and the 4 action inputs. The only 1 output is the action value. According to the expected control objective, the reward function at time t_k is set as

$$\begin{aligned} r_k(\boldsymbol{x},\boldsymbol{u}) = & w_h|h_k - h_c| + w_{vx}|(v_x)_k - v_{xc}| + w_{vy}|(v_y)_k - v_{yc}| \\ & + w_{vz}|(v_z)_k| + w_\beta|\beta| + w_\phi|\phi_k| + w_\psi|\psi_k - \arctan(v_{yc}/v_{xc})| \\ & + w_p|p_k| + w_q|q_k| + w_r|r_k| \\ & + w_{ih}\left|\int(h_k - h_c)\mathrm{d}t\right| + w_{ivx}\left|\int((v_x)_k - v_{xc})\mathrm{d}t\right| + w_{ivy}\left|\int((v_y)_k - v_{yc})\mathrm{d}t\right| \end{aligned} \tag{18}$$

where $w_i\ (i = h, v_x, v_y, v_z, \beta, \phi, \psi, p, q, r, ih, iv_x, iv_y)$ are weight coefficients. In learning the control law, an episode is defined as a finite horizon including a certain number of control periods of ΔT. The discount factor is γ=0.99, and then the return function is $R = \sum \gamma^k r_k$.

Use Pytorch to build the NN model. The structure of the Actor network is the similar to that of Sec. 3.2, except the input layer. The Critic network has 4 layers. There are 2 hidden layers, with the number of units being 64 and 128, respectively. For the first 3 layers, the activation function is "Relu" function, while the activation function is linear for the output layer. Both networks are initialized with the "xavier_uniform" function. The mini-batch size is N=128 and the length of an episode is 5000 ΔT, with ΔT=0.1s. In addition, if the aircraft states exceed some preset boundary value, the episode will terminate immediately with penalty on the rewards. The weight coefficients in the reward function (18) are $w_i = 0.01 (i = h)$, $w_i = 0.01 (i = v_x, v_y, v_z)$, $w_i = 2 (i = \beta, \phi, \psi)$, $w_i = 1 (i = p, q, r)$, and $w_i = 0.05 (i = ih, iv_x, iv_y)$. After an episode ends, the Critic network and Actor network parameters are updated with the "Adam" optimization method. The learning rate are taken as $\alpha = 1\times10^{-3}$ and $\beta = 5\times10^{-3}$, respectively, and then the soft update for the target networks are performed, with a update rate of $\tau = 0.001$.

The Python platform "Spyder" is employed to carry out the training of the NN controller. The training results are presented in Figs. 4-6. In Fig. 4, the average return function against the number of episodes is plotted. The length of episodes is given in Fig. 5 and the smoothed terminal error during the training is given in Fig. 6. It may be found that the performance of the NN controller is not improved monotonically. However, as the number of the training increases, the results show an overall improvement on the number of times when the aircraft successfully finishes the episodes. Compared with the quadrotor, the reinforcement learning control of the fixed-wing aircraft is more complicated, due to the balance between the aerodynamic force and aerodynamic moment. Moreover, for the full-sized aircraft, the training time is also greatly increased compared with a small aircraft.

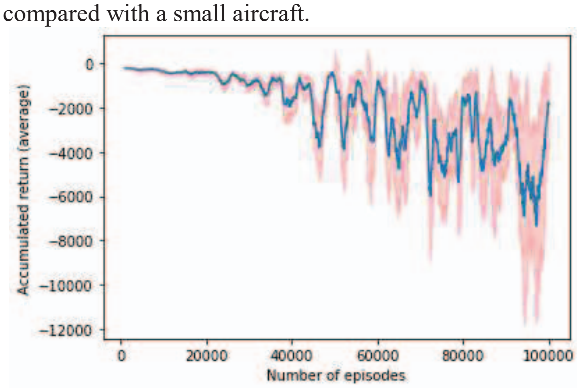

Fig. 4 Accumulated return against number of episodes in the reinforcement learning.

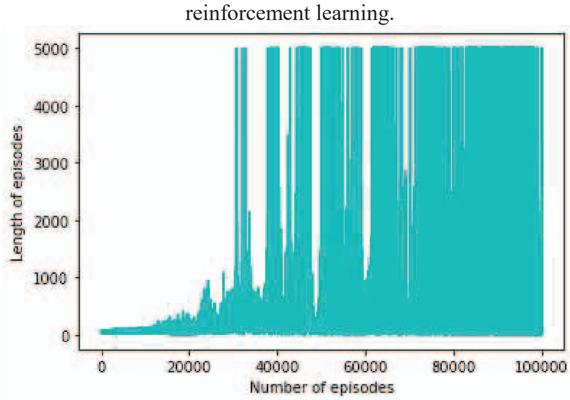

Fig. 5 Length of episodes against number of episodes in the reinforcement learning.

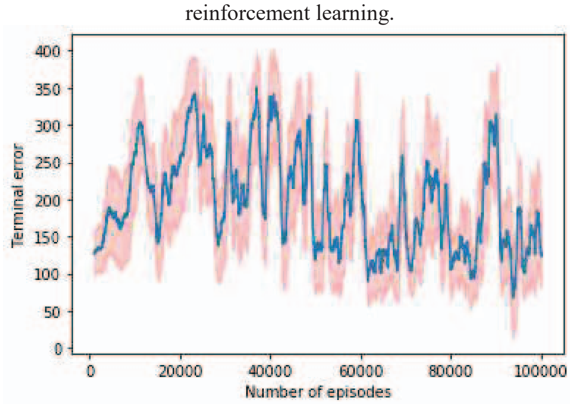

Fig. 6 Terminal error against number of episodes in the reinforcement learning.

5 FLIGHT SIMULATION

Since the monotonous improvement on the performance of the NN controller cannot be guaranteed in the reinforcement learning, it may occur that the control effect of the trained NN controller deteriorates or the controller even destabilizes the aircraft as the training number increases. To get the NN controller that has desirable performance, the terminal errors relative to the target states were compared after each training episode and the NN controller with smaller error was selected.

Upon the NN controller obtained after 78890 episodes of training, the flight simulation, with a time horizon of 1000s, was carried out on "Spyder". The initial conditions and the target states are the same as those in Sec. 3. Figure 7 shows the horizontal trajectory of the aircraft, and the aircraft realizes the commanded heading. Figure 8 gives the height of the aircraft. It is shown that the height of the aircraft converges to the commanded value of 100m after 50s and the terminal error is 0m. The velocity profiles of the aircraft are given in Fig. 9. After a transition period of 40s, the aircraft approaches the commanded velocity and the commanded values are achieved exactly. Figure 10 plots the attitude Euler angles during the flight. It is shown that there is a steady roll angle of -5.84° during the cruise flight. The angular velocities of the aircraft are plotted in Fig. 12, and they all tend to zero after 40s.

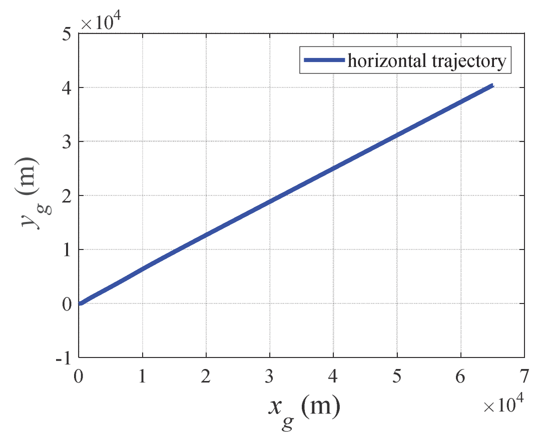

Fig. 7 Aircraft horizontal trajectory under the reinforcement learning NN controller.

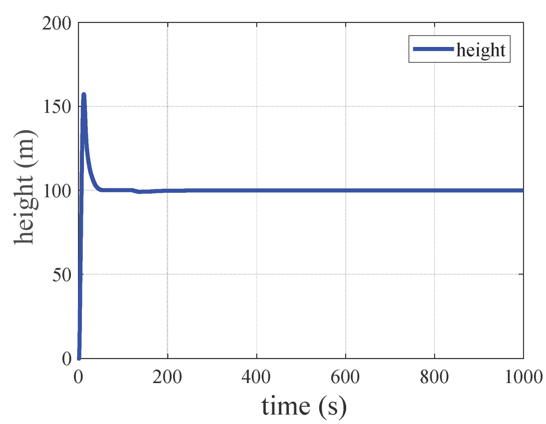

Fig. 8 Aircraft height under the reinforcement learning NN controller.

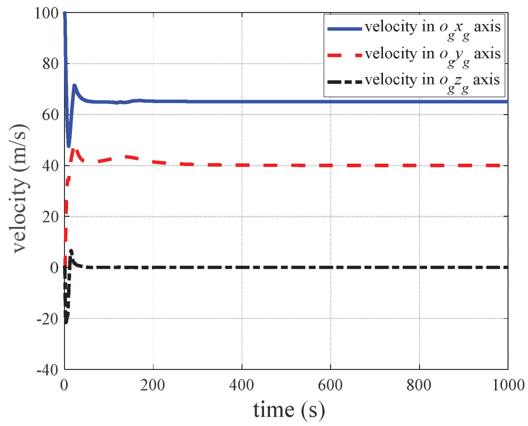

Fig. 9 Aircraft velocity under the reinforcement learning NN controller.

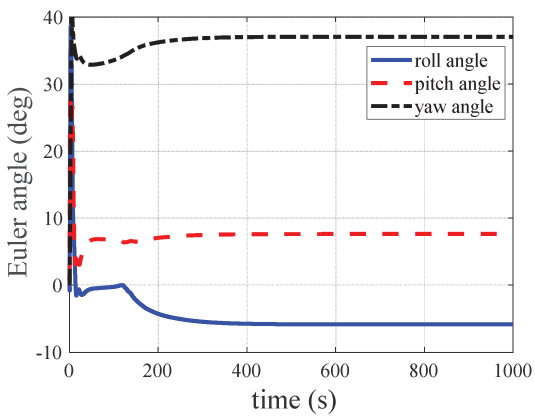

Fig. 10 Aircraft attitude under the reinforcement learning NN controller.

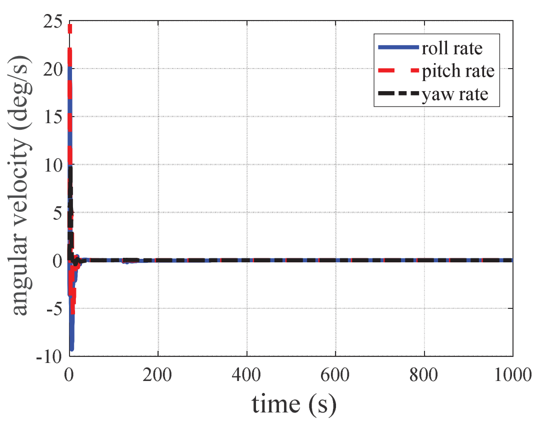

Fig. 11 Aircraft angular velocity under the reinforcement learning NN controller.

In Fig. 12, the aircraft angle-of-attack and sideslip angle are presented. Different from the height and the velocity control, which reach the commanded value exactly. It is shown that the aircraft still maintains certain sideslip of -6.21° after entering the stable flight. Figure 13 gives the aero-surfaces deflection and throttle control of the aircraft. Since the reward function does not consider the penalty on the control inputs, the control transition process of the fixed-wing aircraft is not satisfactory. However, except the drastic changes at the initial stage, the control outputs all reach the equilibrium value gradually. To sum up, under the control of NN controller, despite of the fixed sideslip and roll, the aircraft enters a stable cruise mode and the control objective is achieved.

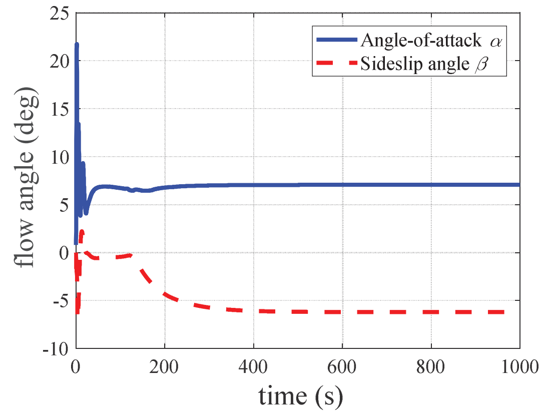

Fig. 12 The angle-of-attack and sideslip angle under the reinforcement learning NN controller.

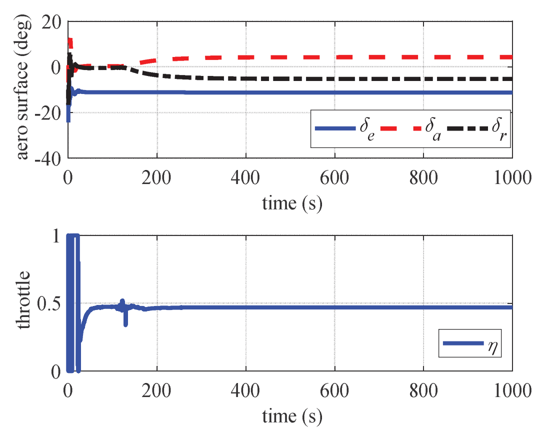

Fig. 13 The aero-surfaces and throttle control under the reinforcement learning NN controller.

6 CONCLUSION

Applying Neural Networks (NNs) for controller design is not novel. However, for the NN controllers studied before the Deep Reinforcement Learning (DRL), the NNs are mainly combined with traditional controllers to approximate the plant model, or used to control the plants of special inverse structure. In essence, those two manners approximate the system model, while with the DRL, the control law may be directly learned upon the generalization ability of NN models. Through this learning way, the end-to-end integrated control for the 6 degree-of-freedom flight of the fixed-wing aircraft is studied in this paper. Compared with the quadrotor, the implementation of intelligent control on the fixed-wing aircraft is more complicated since it needs to balance the aerodynamic force and the aerodynamic moment. To our encouragement, via the reinforcement learning, it is shown that the trained NN controller can learn the control law and achieve the control objective. However, the control quality and generality of the NN controller still need to be improved, and these issues will be studied in the future.

REFERENCES

[1] B. Stevens, F. L. Lewis, E. N. Johnson, Aircraft control and simulation: dynamics, controls design, and autonomous systems, Wiley-Blackwell, 2015.

[2] S. A. Snell, D. F. Enns, W. L. Garrard. Nonlinear inversion flight control for a supermaneuverable aircraft, Journal of Guidance Control and Dynamics, Vol. 15, No. 4, 976-984, 1992.

[3] J. Farrell, M. Sharma, M. Polycarpou, Backstepping based flight control with adaptive function approximation, Journal of Guidance Control and Dynamics, Vol. 28, No. 6, 1089-1102, 2005.

[4] S. Zhang, Q. Wang, K. F. He, Y. P. Shao, An improved dynamic surface control law and its application in post-stall maneuver, Acta aerodynamica sinica, Vol. 35, No. 5, 718-726, 2017.

[5] I. M. Gregory, C. Y. Cao, E. Xargay, N. Hovakimyan, X. T. Zou, L1 adaptive control design for NASA AirSTAR flight test vehicle, AIAA Guidance, Navigation, and Control Conference, AIAA-2009-5738, 2009.

[6] Q. Wang, W. Q. Qian, D. Ding A review of unsteady aerodynamic modeling of aircrafts at high angles of attack, Acta Aeronautica et Astronautica Sinica, Vol. 37, No. 8, 2331-2347, 2016.

[7] L. B. Xu, Introduction to Machine Learning, Changchun: Jilin University Press, 1993.

[8] R. Sutton, A. Barto, R. Williams, Reinforcement learning is direct adaptive optimal control, IEEE Control Systems Magazine, Vol. 12, No. 2, 1-22, 1991.

[9] R. Sutton, A. Barto, Reinforcement Learning: An Introduction, the MIT Press, Massachusetts, 2018.

[10] W. Wei, Learning Control and Its Application in Robot Control, Hangzhou: Ph.D dissertation of Zhejiang University, 1994.

[11] X. Xu, Reinforcement Learning and Its Applications in Navigation and Control of Mobile Robots, Changsha: Ph.D dissertation of National university of defense science and technology, 2002.

[12] W. L. Cai, Research on Control Method for Micro Unmanned Helicopter via Reinforcement Learning, Changsha: Master's thesis of National university of defense science and technology, 2007.

[13] Q. H. Zhu, The Application of BP Multi-Layer Neural Network in Control, Nanning: Master's thesis of Guangxi University, 2004.

[14] L. Cao, Satellite Orbit Prediction Method Based on Compensation Model by Using Deep Neural Network, Nanjing: Master's thesis of Nanjing University of Aeronautics and Astronautics, 2014.

[15] V. Mnih, K. Kavukcuoglu, D. Silver, et al., Human-level control through deep reinforcement learning, Nature, Vol. 518, 529-533, 2015.

[16] T. P. Lillicrap, J. J. Hunt, A. Pritzel, N. Heess, et al., Continuous control with deep reinforcement learning, arXiv preprint. arXiv: 1509.02971v5[cs.LG], 2016.

[17] S. Zhang, J. Xiao, J. T. Huang, K. F. He, Intelligent flight control for quadcopter based on the deep reinforcement learning, the 2020 Chinese Conference on Flight Mechanics and Flight Test, 2020.

[18] D. S. Naidu, A. J. Calise, Singular perturbations and time scales in guidance and control of aerospace systems; a survey, J Guid Control Dynam, Vol. 24, No. 6, 1057-1078, 2001.

[19] S. G. Clarke, I. Hwang, Deep reinforcement learning control for aerobatic maneuvering of agile fixed-wing aircraft, AIAA Scitech 2020 Forum, Orlando, FL, 2020.

[20] S. T. Wu, Y. H. Fei, Flight Control System, Beijin: Beijing University of Aeronautics and Astronautics Press, 2005.

[21] A. Chakraborty, P. Seiler, G. J. Balas, Applications of linear and nonlinear robustness analysis techniques to the F/A-18 flight control laws. AIAA Guidance, Navigation, and Control Conference, Chicago, Illinois. AIAA 2009-5675, 2009.

[22] D. Kingma, J. Ba, Adam: A method for stochastic optimization, arXiv preprint, arXiv: 1412.6980. 261, 2014.

[23] Y. L. Bai, Research and Improvement on Deep Neural Network Based Object Recognition, Harbin: Master's thesis of Harbin Institute of technology, 2014.